Georgia State University
CSC 4210/6210 Computer Architecture
Spring Semester 2026

Processor Design Semester Project - Task2:

Release Date: 03-10-2026
Due Date: 03-20-2026

Truth Table → Boolean Equation → K-Map Simplification

You are designing combinational logic as part of a processor datapath. In real hardware design, logic often begins as a truth table which must be converted into a Boolean equation and simplified for efficient implementation.

Section 1 — Input System

1. Allow user argument to specify the number of input variables ($n \geq 2$)
2. Accept a truth table provided by the user
3. You may design any input method (console input, file input, interactive prompt, etc.)
4. Validate the truth table:
    a. The table must contain 2^n rows
    b. Each input combination must appear exactly once
    c. Output values must be 0 or 1

Section 2 — Boolean Expression & Simplification

1. Convert the truth table into a Boolean equation:
    a. The user must be able to select one of the following forms:
        i. SOP (Sum of Products), POS (Product of Sums)
        ii. Optionally you have choice to choose from SOP/POS
    b. The program must:
        i. Generate the canonical equation (SOP or POS)
        ii. Optionally you have choice to choose from SOP/POS
        iii. List the corresponding minterms or maxterms
2. Simplify the Boolean equation
    a. For functions with 2–4 variables, construct a Karnaugh Map (K-Map) and perform grouping to derive the simplified Boolean expression.

Section 3 — Validation

1. Validate the simplified expression by evaluating it for all input combinations and comparing it with the original truth table

Program Output:

1. Truth table
2. Canonical equation (SOP or POS)
3. Minterm / Maxterm list
4. K-Map grouping
5. Simplified Boolean expression
6. Validation result (PASS / FAIL)

Deliverables:

1. Source code (Python preferred)
2. README with instructions
3. Submit code to github
4. 1-2 minute demo video explaining code and displaying input/output
    a. Either upload it to iCollege or storage of your choice with necessary viewing permissions
5. Can video recordings be uploaded to youtube? Yes or No.